

Program overview

14-Dec-2022 11:25

Year 2022/2023
Organization Technology, Policy and Management
Education Minors Management of Technology

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Minors MOT 2022			
MOT-Mi-088-22	MOT-Mi-088-22 Technology-based Entrepreneurship		
TBM016A	Essentials of Technology based Entrepreneurship	4	
TBM017A	Case Study in Technopreneurship	10	
TBM020C	Managing Start-ups: teamwork, leadership and AI	3	
TBM026A	Introduction to Technology Based Entrepreneurship	2	
TBM040A	Product-Service Design & Prototyping	4	
TBM041A	Business Marketing for Engineers	3	
TBM042A	Finance for Entrepreneurs	4	
MOT-Mi-153-22	MOT-Mi-153-22 MedTech Based Entrepreneurship		
IO3880-1	Medical Design 1	2	
IO3880-2	Medical Design 2	2	
TBM016A	Essentials of Technology based Entrepreneurship	4	
TBM018A	Marketing for Start-ups	3	
TBM021B	Introduction to MedTech Entrepreneurship	2	
TBM032A	Health System Management 1	3	
TBM033A	Health System Management 2	2	
TBM042A	Finance for Entrepreneurs	4	
TBM043A	MedTech Integration project 1	4	
TBM044A	MedTech Integration project 2	4	

Year	2022/2023
Organization	Technology, Policy and Management
Education	Minors Management of Technology

Minors MOT 2022

Year	2022/2023
Organization	Technology, Policy and Management
Education	Minors Management of Technology

MOT-Mi-088-22	
Minor Coordinator	Dr. T.L. Dolkens
Administration by the Faculty of	TBM
EC Minor (Volume)	30
Maximum number of participants	100
Minimum number of participants	100
Minor Remarks / Schedule	This Minor is given in Dutch.

TBM016A	Essentials of Technology based Entrepreneurship	4
Module Manager	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Summary	In this course we focus on essential elements when starting a technology-based startup. The aim is to understand the commercial potential of implementing new technology and how to organize a startup to bring the product, process or service into the market. You will learn about technology-based innovations and business model innovations to bring new products and services to the market. During the course, we touch on various items, considered to be essential for technology-based entrepreneurship such as intellectual property rights, technology development, strategic market and value chain positioning and the role of entrepreneurial ecosystems to support the growth of newly founded startups. You will learn about the process of opportunity identification and the dynamics of early growth of new firms. Entrepreneurs of starting companies are highly dependent on the information they acquire among potential customers and external stakeholders such as venture capitalists, banks, suppliers and partner firms. We will discuss how entrepreneurs operate in these networks, take benefit from these networks and how they make decisions in such networks. In particular, start-up firms are likely to collaborate with larger incumbent firms which have a very different approach to the management of their business as compared to start-ups. Knowing the difference and how to operate in such instances makes the entrepreneur more resilient and successful. Finally, you will learn about the internal organization of the firm and team development during stages of growth. Important element of the course is critical thinking. Entrepreneurship is very diverse, manifests in various forms and situations. The variety of entrepreneurship is impossible to capture in a single model, concept or tool. These models, concepts and tools have great benefits for certain forms and situations of entrepreneurship but lack usefulness in other. To be successful as an entrepreneur, it is key to understand when models, concepts and tools make sense.	
Course Contents	Identifying opportunities for technology-based business: Creation versus discovery. Finding applications for new technologies; Maturity analyses of technologies and markets; Appropriability: how to create, deliver and capture value; Innovation models; Inventive problem solving; Intellectual property management; Choosing a business model strategy; From product idea to technostarter.	
Study Goals	After completing this course, you will be able - to understand and recognise what technolog-based entrepreneurship is; - to analyse entrepreneurial opportunities, how they emerge and how they offer value to your customers and to you to run a sustainable and repeatable business model; - to analyse the complexity of start-up growth and the role of markets, stakeholders and the entrepreneurial ecosystem and be able to analyse how these affect the growth of the new firm; - to analyse the context of technology innovation and the organizational form and business model to deal with risks and opportunities associated with technology-based entrepreneurship. - to analyse the core value of your business in the eyes of the customer and how to protect this from others doing the same. Entrepreneurship can manifest in many forms and no single tool applies best. Therefore it is important that you will develop a critical view on the concepts, methods and tools that are available for technostarters. Many tools and methods do exist and it is key to know which applies in your situation. This critical view will be assessed.	
Education Method	Lectures will be partly online and in small class setting. The course is organized in seven sessions. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments. During the course you need to study the literature and you will work on a small case study assignment during the lectures within a team of two students.	
Assessment	Offline lectures will focus on class discussion, in which we will further deepen our understanding of the topics of the class. The course is graded based on two components 1. Two team assignments (40%) 2. An individual exam (60%). In normal circumstances, the exam is a closed-book (in-person, on campus) open question and multiple-choice examination of 3 hours. In case of unforeseen circumstances or due to measures resulting from COVID-19, the exam will be given in the form of an on-line (digital) open-book & timed open question and multiple-choice examination of 3 hours.	
Targetgroup	The course is only available to students that take part in a minor Entrepreneurship program, e.g. Minor MOT-Mi-088 and MOT-Mi-153.	

TBM017A	Case Study in Technopreneurship	10												
Module Manager	Dr. T.L. Dolkens													
Co-responsible for assignments	Dr.ing. V.E. Scholten													
Contact Hours / Week x/x/x/x	4/4/0/0													
Education Period	1 2													
Start Education	1													
Exam Period	2													
Course Language	English													
Course Contents	De Case Beschrijving van een Technologische Onderneming (TBM017A) vormt het integratiemoment van de Minor Technology-Based Entrepreneurship (MOT-MI-088-19), waarin de geleerde kennis en vaardigheden uit de individuele minorvakken (modules) samenkomen. Niets is namelijk zo instructief als het zelf aan de slag gaan om alle geleerde kennis ook daadwerkelijk toe te passen. Het doel is om een (eigen) innovatief idee uit te werken en te ontwikkelen naar een realistische marktpropositie. Ieder team, bestaande uit 5 studenten vanuit verschillende studierichtingen, komt zelf tot een innovatief productidee en werkt dat aan de hand van opdrachten en voorbeelden uit. Leidraad bij dit vak is de Disciplined Entrepreneurship methode van Bill Aulet. Dit tekstboek is verplicht en zal van kaft-tot-kaft worden gebruikt. Het is raadzaam om de papieren versie van dit boek te kopen, zodat je er snel in kunt bladeren en op relevante plekken Post-Its kunt plakken en aantekeningen kunt maken. Het is een nuttige aanschaf want je zult er vaak op terugvallen. Wat dat betreft is een e-book geen goed alternatief. Het integratievak wordt afgesloten met het opleveren en presenteren van een business plan waarin het idee staat beschreven, maar vooral de manier waarop het in een behoefte voorziet en hoe je daaraan tegemoet gaat komen. Een heel innovatief idee is leuk, maar als niemand er op zit te wachten, is het waardeloos. De essentie is dat je overtuigend kunt aangeven hoe jullie idee op een lucratieve manier in de markt kan worden gebracht, rekening houdend met aspecten als patenten, financiën, productontwikkeling, marketing, etc. De individuele modules van de Minor worden als flankerend onderwijs toegepast binnen deze casebeschrijving. Realiseer je dat, zelfs als je de beste muizenval ter wereld bouwt, de mensen die last hebben van muizen niet automatisch je deur plat lopen. Daar moet je wat voor doen. Wat precies? Dat leer je in deze Minor.													
Study Goals	Doelstelling van deze course: het geïntegreerd toepassen van kennis en vaardigheden op het terrein van ondernemerschap op basis van technologie. Na afronding van de Minor ben je voorbereid op ondernemerschap in de meest brede zin van het woord. Je hebt de kennis en vaardigheden opgedaan die nodig zijn voor het starten van een eigen bedrijf en je bent in staat ondernemend te handelen en initiatieven te ontplooiën binnen bestaande organisaties. Je ontwikkelt de vaardigheden om een technologische onderneming te starten en de markt waarin zij opereert te analyseren; Je leert (iteratief) steeds beter om de relevante product-markt combinaties te ontwikkelen; Je leert hoe je de gekozen product-markt combinatie kan implementeren in de bestaande markt en je maakt een gedegen concurrentieanalyse; Je vindt het businessmodel dat het best past bij jouw innovatieve oplossing. Soms is het businessmodel zélf een belangrijk deel van de innovatie.													
Education Method	Zelfs als je later geen eigen onderneming gaat starten, heb je in deze minor de kennis en vaardigheden opgedaan om een project te initiëren, of bij een bestaand bedrijf een nieuwe business unit op te zetten. Je ontwikkelt een innovatief product en gaat dat in de praktijk toetsen. Dat betekent dat je in een vroeg stadium van de Minor naar buiten gaat, en met mensen gaat praten. Talking to other carbon-based life forms blijkt een van de moeilijkste opgaven voor een ingenieur. Je moet gaan praten met mensen waarvan je denkt dat het je toekomstige klanten worden, en met mensen die je kunnen helpen om je idee te realiseren. Het uiteindelijke doel (en een van de deliverables voor dit integratievak) is een goed uitgewerkt business plan met zo weinig mogelijk aannames en onzekerheden. Gedurende de course vindt een aantal bijeenkomsten plaats waar je met je team de bevindingen van de verschillende opdrachten presenteert en onderbouwt. In plenaire colleges worden de opdrachten door de docenten uitgelegd en geïllustreerd aan de hand van voorbeelden en gastsprekers. Halverwege het semester presenteert je met je team het conceptvoorstel voor een nieuwe product-markt combinatie. Als dit resulteert in groen licht werk je dit idee met je team verder uit, en schrijft een gedegen eindrapportage (business plan) over de implementatie van de voorgestelde innovatieve product-markt combinatie.													
Assessment	Het eindcijfer van de cursus is als volgt opgebouwd: <table><tr><td colspan="2">Onderdeel</td></tr><tr><td>1. Individuele participatie (aanwezigheid, participatie en peer score):</td><td>20%</td></tr><tr><td>2. Deelopdrachten (5) in oktober, november en december:</td><td>50%</td></tr><tr><td>3. Final Businessplan (document):</td><td>20%</td></tr><tr><td>4. Presentatie Final Businessplan (voor jury):</td><td>10%</td></tr><tr><td>Totaal</td><td>100%</td></tr></table> Retakes worden nader vastgesteld, indien nodig. De deelopdrachten en de eindrapportage worden in groepsverband uitgevoerd en zijn de basis voor het groeps cijfer. De participatie tijdens de bijeenkomsten wordt individueel beoordeeld en verdisconteerd in de individuele eindcijfers. Daarnaast zal ieder team een interne evaluatie doen waarin ieder groeps lid de inzet, motivatie en team spirit van de alle groepsleden (inclusief zichzelf) zal beoordelen. Deze evaluatie wordt ook in rekening gebracht op de uiteindelijke individuele cijfers. Wees niet verbaasd als de eindcijfers van twee leden van dezelfde groep flink uit elkaar liggen. Het is dus zaak je vanaf het begin volledig in te zetten voor je groep, want het groeps cijfer is slechts de basis voor de individuele eindcijfers. Voor alle onderdelen van het vak geldt een drempel van 5,5. De onderlinge samenhang tussen de toets- onderdelen zal op het eerste college worden toegelicht. Vanwege de dan actuele Covid-19 restricties zijn colleges en examens mogelijk online.		Onderdeel		1. Individuele participatie (aanwezigheid, participatie en peer score):	20%	2. Deelopdrachten (5) in oktober, november en december:	50%	3. Final Businessplan (document):	20%	4. Presentatie Final Businessplan (voor jury):	10%	Totaal	100%
Onderdeel														
1. Individuele participatie (aanwezigheid, participatie en peer score):	20%													
2. Deelopdrachten (5) in oktober, november en december:	50%													
3. Final Businessplan (document):	20%													
4. Presentatie Final Businessplan (voor jury):	10%													
Totaal	100%													

TBM020C	Managing Start-ups: teamwork, leadership and AI	3
Module Manager	B. Wagner	
Instructor	M.T. Sekwenz	
Instructor	Dr. G. de Vries	
Instructor	Dr. H.G. van der Voort	
Instructor	B. Wagner	
Instructor	Mr. J.M. Kooijman	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course is only open to students enrolled in the Technology-Based Entrepreneurship minor.	
Course Contents	When technological start-ups launch a product, process or service in the market, they have to deal with various management issues. For instance, issues concerning organisational structure, human resource management (HRM), labour law, teamwork, leadership and power. During this course, students learn about the specific risks and opportunities startups face in management, compared to more traditional organisations. Students also learn about AI recruitment and AI HRM tools for startups, as well as their strengths and weaknesses. Students learn how to reduce potential risks and seize the opportunities that good human resources management offers.	
Study Goals	1. Students can distinguish and recognize different types of organizational structures. They can describe the risks and opportunities of startups compared to other (types of) organisations. 2. Students are able to describe the risks and opportunities of startups with respect to human resource management (e.g., attracting, retaining and dismissing employees). They can also formulate strategies to reduce the HRM risks and seize opportunities. 3. Students can describe the group processes that can occur in startups and what the risks and opportunities of these processes are for staff performance. They can formulate strategies to reduce these risks and exploit the opportunities. 4. Students are able to identify leadership styles and to name the advantages and disadvantages of these styles. 5. Students are able to classify and identify the different types of power most important for human resource management. 6. Students are able to identify AI tools used for recruitment and HRM of startups and identify the strengths and weaknesses of these tools.	
Education Method	Lecturers	
Assessment	Individual online exam and groups essay	

TBM026A	Introduction to Technology Based Entrepreneurship	2
Module Manager	Dr. T.L. Dolkens	
Instructor	Ir. N. Stoimenova	
Instructor	E. Çelik	
Responsible for assignments	Dr. T.L. Dolkens	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none Different, to be announced	
Course Language	English	
Required for	The course is only available to students that take part in a minor Entrepreneurship program, e.g. Minor MOT-Mi-088.	
Course Contents	Many entrepreneurship programs are derived from management programs and are intended to teach managers how to become more entrepreneurial, yet they are not aimed at the entrepreneur as such (Katz, 2007). For example, in business schools the focus is much on 'the' ingredient of entrepreneurship: a business plan presented through a series of case studies and lectures (Carrier, 2007, p. 143). While understanding how big organizations and managers think about entrepreneurship is important (the last two recommended books below are focused on that), in this course we will focus on what it means to be a self-employed entrepreneur, someone who takes the risks on personal account. We will focus on what it takes to become a successful entrepreneur, what entrepreneurial activity is and why teams of entrepreneurs are considered more resilient in the face of uncertainty.	
Study Goals	You will further develop your competence for self-directed and entrepreneurial learning. You'll learn to see and initiate new opportunities in an imaginative way within the Minor TBE. You provide evidence of the development of your entrepreneurial competences by means of an orally delivered coherent and gripping story that is supported by annotated documentation of your learning attempts. You can give useful and to-the-point feedback on the portfolios and stories of your team members or other fellow students. You can reformulate/reframe the feedback you have received either by lecturers or other fellow students into meaningful new learning objectives for the Minor and the semester thereafter. You become a highly functional part of your team by i.e. recognizing your own role within that team and proposing well-founded interventions to improve your work and team dynamics.	
Education Method	Online lectures; online exercises; case studies; guest speakers. The teacher(s) will coach you on an individual and team level throughout this elective. The emphasis of teaching is in Quarter 1 of the Minor, but this subject will partly continue in Quarter 2. You will receive concise written and oral formative feedback to help you achieve the goals you've set at the beginning. To achieve all that, you are expected to develop a portfolio of your entrepreneurial learning and continuously reflect on it (and update it) throughout Quarter 1 and part of Quarter 2. You then submit a final portfolio at the beginning of Q2 when your development will be tested during an oral examination. The submitted portfolio serves as an underlying 'evidence' of your development. We use the integral case of the Minor as content and context for your attempts at 'learning to be an entrepreneur'. We also discuss opportunities outside of the curriculum, e.g. in your own company, sports/student/culture/student association, your study, your choice of Master, etc. Entrepreneurial learning is not limited to the 'walls of the university'.	
Assessment	The course is graded based on two components that are done as take-home assignments and are considering the restrictions due to Covid-19 1. Two team assignments of take-home case work in the first week (40%) 2. Individual reflection on team competences and your own role (20%) 2. An individual report reflecting on individual entrepreneurial competences (40%). All 3 exam components are online, due to Covid restrictions.	

TBM040A	Product-Service Design & Prototyping	4
Module Manager	Dr. T.L. Dolkens	
Instructor	P. Harish	
Instructor	Dr. T.L. Dolkens	
Instructor	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	We will follow the book "Testing Business Ideas: Field Guide for Rapid Experimentation by Alexander Osterwalder and David J Bland". We will also make use of various methods, tools and techniques to aid in your product/service development (such as Assumptions Table, Morphological Analysis, Risk Analysis etc.)	
Study Goals	- To be able to define product/service specifications with design requirements and user requirements - To be able to conduct primary market validation with users and other stakeholders (interviews, observations etc.) - To be able to design and prototype various concepts for product/service ideas - To be able to develop a minimal viable product that is ready for traction testing - To be able to assess the minimal viable product for quality assurance and therefore, conduct risk analysis - To be able to design a pilot study/test to validate the product/service	
Education Method	Students work in teams that are formed during Case Study in Technopreneurship course. The lectures are classified into 4 themes namely: Introduction, Problem definition & Ideate, Prototype and Validate. Each theme has at utmost 2 lectures, where each lecture will have ONE hour of topics discussed and another ONE hour of hands-on activities. Student teams will also receive advice during coaching sessions from the instructor. During the course, various assignments are given that will be discussed and presented in the class. The course ends with a prototype developed and a final presentation (in the form of an exposition).	
Assessment	The course has 3 grading components : - Weekly in-class assignments - Jury assessment during the final presentation - Prototype	

TBM041A	Business Marketing for Engineers	3
Module Manager	Dr. T.L. Dolkens	
Instructor	Dr. T.L. Dolkens	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	This an introductory course (for engineers) on the basic principles of Business to Business Marketing in an engineering, entrepreneurial environment.	
	The primary aim of the course introduction to business marketing is to develop interest in, become acquainted to and encourage application of fundamental concepts, models and instruments in the area of marketing. It provides an introduction to specific knowledge of marketing as a function in organizations and places this in a entrepreneurial. In addition to that, it aims to provide insights into current developments, newest ideas and largest challenges in this field.	
	The course aims at understanding the basics of Business to Business Marketing, more explicit in understanding markets, clients, purchasing behaviour, market research and forecasting, marketing planning, prices setting and industrial marketing communication.	
	This course will be taught in an application-oriented way. You have to make this course by yourself (this is called Learner Led Learner of reversed teaching). You master your own learning path. We want to achieve this by case studies and presentations. Interactivity is a keyword in this course. If you do not interact, you will not learn.	
Study Goals	The various B2B marketing concepts and principles will be discussed during the lectures. To become acquainted with these concepts, the students will work on several theoretical cases. To get graded you will have to perform in groups (strategic marketing plan).	
	Learning Objectives: After this course you have gained knowledge about the way organizations market their goods and services, the mechanism of marketing, the activities that belong to sales, the effects on relationships in the supply chain, marketing processes and the effect of e-business on the processes and supply chain.	
	After this course you should be able to: - Identify the relevance of marketing - Analyze and identify improvement opportunities in the basic marketing process and organization - Identify, segment and select customers, differentiating on the basis of their importance to the firm - Identify and use relevant measurements for measuring marketing and customer performance	
	two hours lecture a week, on tuesdays.	
Education Method	two hours lecture a week, on tuesdays.	
Literature and Study Materials	- Lecture sheets.	
	- Selected academic articles on Brightspace - Reprints of: Dwyer, Robert F., and Tanner, John F., Jr., (2002), Business Marketing: Connecting Strategy, Relationships, and Learning, Second Edition, McGraw-Hill Irwin, Burr Ridge, IL., ISBN 0-07-112332-6 (not available in print anymore)	
Assessment	The course will be graded through a group assignment, ie a strategic marketing plan.	

TBM042A	Finance for Entrepreneurs	4
Module Manager	Dr. T.L. Dolkens	
Instructor	Dr. A. Giga	
Co-responsible for assignments	A. Boru	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is necessary. Knowledge of basic finance and economics helps.	
Course Contents	This course focuses on startup financing. We cover finance tools and concepts that are essential to an entrepreneur when creating a financing strategy. We go over the techniques for setting up and evaluating a financial model and cover the principles of sound financial management. We study fundraising strategy and the entrepreneurial finance industry: the sources of startup funding (e.g. Venture Capital), motivations of entrepreneurs and investors, and negotiations and contracts between them. We also explore latest developments in entrepreneurial finance (e.g. crowdfunding).	
Study Goals	(1) learn how to be a good steward of your business and critically assess financial viability of a business model; (2) know where you can obtain funds for your business, the trade-offs with each funding source, and how it fits your business model; (3) become well-versed in fundraising strategy; (4) master or review fundamental concepts in finance.	
Education Method	The material is illustrated through a combination of lectures, case studies, in-class exercises, and discussions. Depending on the availability and scheduling, we may host guest speakers, either entrepreneurs or investors, to share their experience directly related to the course material.	
Literature and Study Materials	To be announced at the start of the module.	
Assessment	Group project related to the minor project (30%), problem sets (30%), written exam (40%). In case of unforeseen circumstances or due to measures resulting from COVID-19, the exam will take form of a timed take-home exam administered online	
Remarks	If you're not concurrently taking one of the entrepreneurship minors, special permission is needed to attend the class. Please e-mail the instructor for permission.	

Year	2022/2023
Organization	Technology, Policy and Management
Education	Minors Management of Technology

MOT-Mi-153-22

Minor Coordinator	A. Boru
Program Coordinator	A. Boru
In association with the Faculty of	3mE IO TBM
EC Program	30 ECTS
Program Start (Study Year)	1&2 (Starts in Q1 ends at the end of Q2)
Administration by the Faculty of	TBM
Administration by the Education of	Delft Centre for Entrepreneurship (DCE)
Minor Title	MedTech Based Entrepreneurship
Minor Semester	Semester 1
Contact for Students (Minor)	A. Boru
Intended for	Bachelor students (with an entrepreneurial mindset) from all faculties of TU Delft, Leiden and Erasmus Universities. We also have growing number of medical students in our minor program.
Minor Exit Qualifications	Technology is playing an increasingly important role in healthcare. Technological innovations make it increasingly possible in the healthcare sector. It is attractive for an engineer to work in a medical context and it is important that doctors know about about innovating in health care sector. Market forces in health care are currently the dominant approach in terms of harmonizing goals and resources. As a result, the company has become an important link. The minor offers a basis for students who want to get acquainted with entrepreneurship in general and also for students who like to apply their technical knowledge and skills within a medical context. The minor is offered by the Delft Center for Entrepreneurship (DCE) and lecturers from the faculties of TPM, IDE and 3ME. After completing the minor, students have basic knowledge of: * doing business in general, * doing business in a medical context, * the design process for the medical user, * the requirements for certification and standardization of medical products, * the financial system of care and of a company, * the economy and management of care, * and marketing & sales of a medical product.
Minor Coherence / Goal	The subjects that belong to this minor aim to provide basic knowledge and skills of / in entrepreneurship in general and entrepreneurship with a medical technology or product. Each course highlights part of the entrepreneurial process and / or specifically addresses the medical context. General knowledge about entrepreneurship, as well as business models, business economics and marketing comes from the Delft Center for Entrepreneurship (DCE). Knowledge about the design process for a medical target group comes from the faculties of IDE and 3ME (including knowledge about the standardization and certification of medical products). This is supplemented with specific knowledge about the organization and management of healthcare institutions and the economy of the healthcare market (TBM).
Minor Content 1	The courses that belong to this minor aim to give you a good idea of the specific aspects of medtech-based entrepreneurship. Each course highlights one or more aspects of technopreneurship in health care. The complete course package covers the most important aspects of entrepreneurship. Overview courses of this minor: - Introduction to Medtech Entrepreneurship, 2 credits: Understanding medtech entrepreneurship, through examples from start-ups and experiences of guest lecturers. - Health System Management 5 credits (Part-1 3 credits, Part-2 2 credits), The Theory of Care Economic Concepts and Principles. The practice of Dutch health care. - Essentials of Technological Entrepreneurship, 4 credits: The commercial potential of new technology and the organization of a new company to realize this. - Medical Design 4, credits (Part-1 2 credits, Part-2 2 credits): The design process and the end user of medical innovations are central to this module. - Finance for Entrepreneurs, 4 credits: This course focuses on the financial management and assessment of a company. - Marketing for Start-ups, 3 credits: Introductory course on the basic principles of Business to Business Marketing in a technical, medically oriented, entrepreneurial environment. - Integration Project 8 ECTC (Part-1 4 credits, Part-2 4 credits): Setting up your own business or carrying out an assignment in an existing company or start-up in the medtech sector. This course is where you bring all the knowledge and assignments from other courses to create your own company's business model and learn to pitch to the stakeholders of your business.
Education Methods	Lectures, tutorials, work groups and a project.
Minor Assessment	Each course is tested separately. The integrated project will also test the integrated application of the knowledge and skills from the previous modules. Whole minor: exams, reports, business plans and pitching.
Maximum number of participants	60
Remarks Maximum number of participants	We have 5 places for non-TU Delft students
Minimum number of participants	30
Minor Remarks / Schedule	English & Dutch

IO3880-1	Medical Design 1	2
Course Coordinator	Prof.dr.ir. R.H.M. Goossens	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In het vak Medical Engineering leren studenten de theorie van het ontwerpproces voor het ontwikkelen en op de markt brengen van producten voor de gezondheidszorg en passen deze kennis toe om een innovatief product of een innovatieve dienst voor de gezondheidszorg te ontwerpen. Binnen het project wordt medisch product of medische dienst ontworpen. Het doel van het project is oefenen met het toepassen van de theorie uit de hoorcolleges en de vaardigheden die zijn verworven in de workshops. Dit betekent dat studenten samen met patiënten, zorgprofessionals, mantelzorgers en/of andere belanghebbenden samen zullen werken aan het ontwikkelen van een passende oplossing voor het geïdentificeerde probleem. Daarom zijn er drie momenten in het project ingebed waarin gebruikersonderzoek moet worden uitgevoerd om inzicht te krijgen in de behoeften en specifieke problemen van verschillende belanghebbenden en om ontwerpideeën en ontwerpconcept te evalueren en verfijnen.	
Study Goals	Begrijpen en toepassen van basisbegrippen van het ontwerpproces voor producten in het algemeen en medische producten in het bijzonder Opzetten en uitvoeren van exploratief en formatief gebruiksonderzoek en het analyseren van de daaruit voortkomende onderzoeksresultaten. In kaart brengen van de patient journey om inzicht te verkrijgen in de interacties, emoties en barrières voor de patiënt en andere belanghebbenden in het zorgproces. Generen van ontwerpideeën en selectie van kansrijke richtingen voor ontwerpoplossingen. Effectief bijdragen aan teamwork.	
Education Method	In het Medical Design-project ontwerpen studenten het product of de dienst waarvoor ze een bedrijfsplan ontwikkelen in de Integration Course (WM4030TU) van de Minor MedTech Based Entrepreneurship. Studenten leren theorie van het ontwerpproces voor het ontwikkelen van producten voor de gezondheidszorg en passen deze kennis toe bij het ontwerpen van een innovatief product of een innovatieve dienst voor de gezondheidszorg. De cursus omvat een reeks colleges, enkele workshops en een ontwerpproject. Studenten werken aan het ontwerpproject in hetzelfde team van 4-6 personen als in de Integration Course De colleges behandelen theorie en methodes over het ontwerpproces in het algemeen, patient journey mapping en participatief ontwerp voor de gezondheidszorg.	
Assessment	Gedurende het vak werken wordt per team een levende presentatie bijgehouden. Elke week worden daar 2-3 slides aan toegevoegd en 3 maal zal er een presentatie plaats vinden. Alle presentatie hebben een ander publiek, waar de teams hun presentatie op moeten aansluiten. Het eindcijfer is het gewogen gemiddelde van deelvijfers voor de discover, define, develop, deliver fase, samenwerking en inzet. De coaches hebben de mogelijkheid om verschillende cijfers toe te kennen aan individuele studenten.	

IO3880-2	Medical Design 2	2
Course Coordinator	Prof.dr.ir. R.H.M. Goossens	
Contact Hours / Week x/x/x/x		
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In het vak Medical Engineering leren studenten de theorie van het ontwerpproces voor het ontwikkelen en op de markt brengen van producten voor de gezondheidszorg en passen deze kennis toe om een innovatief product of een innovatieve dienst voor de gezondheidszorg te ontwerpen. Binnen het project wordt medisch product of medische dienst ontworpen. Het doel van het project is oefenen met het toepassen van de theorie uit de hoorcolleges en de vaardigheden die zijn verworven in de workshops. Dit betekent dat studenten samen met patiënten, zorgprofessionals, mantelzorgers en/of andere belanghebbenden samen zullen werken aan het ontwikkelen van een passende oplossing voor het geïdentificeerde probleem. Daarom zijn er drie momenten in het project ingebed waarin gebruikersonderzoek moet worden uitgevoerd om inzicht te krijgen in de behoeften en specifieke problemen van verschillende belanghebbenden en om ontwerpideeën en ontwerpconcept te evalueren en verfijnen.	
Study Goals	Uitwerken van ontwerpideeën en selectie van kansrijke richtingen voor ontwerpoplossingen. Ontwikkelen en verfijnen van ontwerpconcepten op basis van inzichten verkregen door middel van gebruiksonderzoek en patient journey mapping, daarbij rekening houdend met relevante organisatorische, ethische en medisch juridische-regulerende randvoorwaarden. Het ontwerpen en vervaardigen van prototypes t.b.v. gebruikersonderzoek Het opzetten en uitvoeren van gebruikersonderzoek en het analyseren van de daaruit voortkomende onderzoeksresultaten. Effectief communiceren van het ontwerpproces naar betrokkenen met beeldend materiaal, schriftelijke rapportage en mondelinge presentaties. Effectief bijdragen aan teamwork.	
Education Method	In het Medical Design-project ontwerpen studenten het product of de dienst waarvoor ze een bedrijfsplan ontwikkelen in de Integration Course (WM4030TU) van de Minor MedTech Based Entrepreneurship. Studenten leren theorie van het ontwerpproces voor het ontwikkelen van producten voor de gezondheidszorg en passen deze kennis toe bij het ontwerpen van een innovatief product of een innovatieve dienst voor de gezondheidszorg. De cursus omvat een reeks colleges, enkele workshops en een ontwerpproject. Studenten werken aan het ontwerpproject in hetzelfde team van 4-6 personen als in de Integration Course De colleges behandelen theorie en methodes over het ontwerpproces in het algemeen, patient journey mapping en participatief ontwerp voor de gezondheidszorg.	
Assessment	Gedurende het vak werken wordt per team een levende presentatie bijgehouden. Elke week worden daar 2-3 slides aan toegevoegd en 3 maal zal er een presentatie plaats vinden. Alle presentatie hebben een ander publiek, waar de teams hun presentatie op moeten aansluiten. Het eindcijfer is het gewogen gemiddelde van deelvijfers voor de discover, define, develop, deliver fase, samenwerking en inzet. De coaches hebben de mogelijkheid om verschillende cijfers toe te kennen aan individuele studenten.	

TBM016A	Essentials of Technology based Entrepreneurship	4
Module Manager	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Summary	In this course we focus on essential elements when starting a technology-based startup. The aim is to understand the commercial potential of implementing new technology and how to organize a startup to bring the product, process or service into the market. You will learn about technology-based innovations and business model innovations to bring new products and services to the market.	
Course Contents	During the course, we touch on various items, considered to be essential for technology-based entrepreneurship such as intellectual property rights, technology development, strategic market and value chain positioning and the role of entrepreneurial ecosystems to support the growth of newly founded startups.	
	You will learn about the process of opportunity identification and the dynamics of early growth of new firms. Entrepreneurs of starting companies are highly dependent on the information they acquire among potential customers and external stakeholders such as venture capitalists, banks, suppliers and partner firms. We will discuss how entrepreneurs operate in these networks, take benefit from these networks and how they make decisions in such networks. In particular, start-up firms are likely to collaborate with larger incumbent firms which have a very different approach to the management of their business as compared to start-ups. Knowing the difference and how to operate in such instances makes the entrepreneur more resilient and successful. Finally, you will learn about the internal organization of the firm and team development during stages of growth.	
	Important element of the course is critical thinking. Entrepreneurship is very diverse, manifests in various forms and situations. The variety of entrepreneurship is impossible to capture in a single model, concept or tool. These models, concepts and tools have great benefits for certain forms and situations of entrepreneurship but lack usefulness in other. To be successful as an entrepreneur, it is key to understand when models, concepts and tools make sense.	
Study Goals	Identifying opportunities for technology-based business: Creation versus discovery.	
	Finding applications for new technologies; Maturity analyses of technologies and markets; Appropriability: how to create, deliver and capture value; Innovation models; Inventive problem solving; Intellectual property management; Choosing a business model strategy; From product idea to technostarter.	
	After completing this course, you will be able	
	to understand and recognise what technolog-based entrepreneurship is;	
	to analyse entrepreneurial opportunities, how they emerge and how they offer value to your customers and to you to run a sustainable and repeatable business model;	
	to analyse the complexity of start-up growth and the role of markets, stakeholders and the entrepreneurial ecosystem and be able to analyse how these affect the growth of the new firm;	
	to analyse the context of technology innovation and the organizational form and business model to deal with risks and opportunities associated with technology-based entrepreneurship.	
	to analyse the core value of your business in the eyes of the customer and how to protect this from others doing the same.	
	Entrepreneurship can manifest in many forms and no single tool applies best. Therefore it is important that you will develop a critical view on the concepts, methods and tools that are available for technostarters. Many tools and methods do exist and it is key to know which applies in your situation. This critical view will be assessed.	
	Lectures will be partly online and in small class setting.	
Education Method	The course is organized in seven sessions. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments. During the course you need to study the literature and you will work on a small case study assignment during the lectures within a team of two students.	
Assessment	Offline lectures will focus on class discussion, in which we will further deepen our understanding of the topics of the class.	
	The course is graded based on two components	
	- Two team assignments (40%) - An individual exam (60%). In normal circumstances, the exam is a closed-book (in-person, on campus) open question and multiple-choice examination of 3 hours. In case of unforeseen circumstances or due to measures resulting from COVID-19, the exam will be given in the form of an on-line (digital) open-book & timed open question and multiple-choice examination of 3 hours.	
Targetgroup	The course is only available to students that take part in a minor Entrepreneurship program, e.g. Minor MOT-Mi-088 and MOT-Mi-153.	

TBM018A	Marketing for Start-ups	3
Module Manager	Dr. T.L. Dolkens	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	not applicable	
Course Contents	TBM018A is an introductory course (for engineers) on the basic principles of Business to Business Marketing in an engineering, entrepreuneuring environment. The primary aim of the course introduction to business marketing is to develop interest in, become acquainted to and encourage application of fundamental concepts, models and instruments in the area of marketing and supply chain management. It provides an introduction to specific knowledge of marketing as a function in organizations and places this in a supply chain perspective. In addition to that, it aims to provide insights into current developments, newest ideas and largest challenges in this field. The course aims at understanding the basics of Business to Business Marketing, more explicit in understanding markets, customers, purchasing behaviour, market research and forecasting, marketing planning, prices setting and industrial marketing communication. This course will be taught in an application-oriented way. You have to make this course by yourself (this is called Learner Led Learner of reversed teaching). You master your own learning path. We want to achieve this by case studies, actual news, internet sites and presentations. Interactivity is a keyword in this course. If you do not interact, you will not learn. The various B2B marketing concepts and principles will be discussed during the lectures. To become acquainted with these concepts, the students will work on several theoretical cases. To get graded you will have to perform both in groups (strategic marketing plan) and individually (exam).	
Study Goals	Learning Objectives: After this course you have gained knowledge about the way organizations market their goods and services, the mechanism of marketing, the activities that belong to sales, the effects on relationships in the supply chain, marketing processes and the effect of e-business on the processes and supply chain. After this course you should be able to: - Identify the relevance of marketing and supply management - Analyze and identify improvement opportunities in the basic marketing process and organization - Identify, segment and select customers, differentiating on the basis of their importance to the firm - Identify and use relevant measurements for measuring marketing and customer performance	
Education Method	two hours lecture a week, on Fridays.	
Books	see below: 'reader'	
Reader	Selected writings from: Dwyer, Robert F., and Tanner, John F., Jr., (2002), Business Marketing: Connecting Strategy, Relationships, and Learning, Second Edition, McGraw-Hill Irwin, Burr Ridge, IL., ISBN 0-07-112332-6 (not available anymore). Readers with articles, links and sheets. The lectures and potential articles are available on the internet site.	
Prerequisites	part of the minor Medtrech	
Assessment	The course will be graded through an exam and a case. The exam is a take home exam, while the case is a group assignment., ie a strategic marketing plan. Both exams are online, due to Covid-19 restrictions.	

TBM021B	Introduction to MedTech Entrepreneurship	2
Module Manager	P.B.M. Vandekerckhove	
Instructor	P.B.M. Vandekerckhove	
Instructor	A. Boru	
Instructor	Dr. A. Giga	
Responsible for assignments	P.B.M. Vandekerckhove	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	18/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Course Contents	The aim of the Introduction to MedTech Entrepreneurship is to inspire and prepare students to use their engineering skills and envision novel technologies in order to solve the challenges that patients and health professionals face. The course is composed of guest lectures given by experts in the health care area. Lectures are scheduled in the first two weeks of classes. Participation to the assigned guest lectures is compulsory. Students are expected to take notes and ask questions during the lectures.	
Study Goals	By the end of the course, students are able to: - Describe the problems in the health care sector. - Discuss and evaluate the diverse perspectives on these problems (by different stakeholders) in health care - Understand how technology plays a role in solving the problems in the health care sector.	
Education Method	Guest lectures and written assignments *Participation to the assigned guest lectures is compulsory.	
Assessment	Written report to be submitted online. Final grade can either be Satisfactory or Unsatisfactory (Voldoende and Onvoldoende)	
Special Information	The scheduling of the talks depends on the availability of the guest lecturers. To accommodate their schedule, some lectures may take place outside of the course's pre-determined timeslot. Students need to remain flexible with their own scheduling during the first two weeks to be able to participate in the guest lectures.	

TBM032A	Health System Management 1	3
Module Manager	Prof.dr. J. Groeneweg	
Instructor	Dr. F.W. Guldenmund	
Instructor	Dr. S. Hinrichs-Krapels	
Instructor	Dr. I. Grossmann	
Co-responsible for assignments	Dr. F.W. Guldenmund	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Course Contents	This course explains what a health system is; who the actors in that system are and how they interact; how knowledge in healthcare develops; and what the role of technology is, now and in the future.	
Study Goals	At the end of the course, participants should be able to 1) Explain what the health system consists of; the institutions involved; applicable laws and regulations; and how these interact. 2) Appraise existing (societal) problems and challenges in the health system or healthcare 3) Discuss scientific methods to improve knowledge and develop evidence-based solutions to health system problems 4) Evaluate the impact of the technological innovations, both on patient care and on the health system, with a special emphasis on aspects of safety.	
Education Method	Lectures blended with tutorial exercises	
Literature and Study Materials	Syllabus health system management 2022	
Assessment	Individual theoretic exam, partially multiple-choice, partially open question	

TBM033A	Health System Management 2	2
Module Manager	Prof.dr. J. Groeneweg	
Instructor	Dr. I. Grossmann	
Instructor	Dr. F.W. Guldenmund	
Co-responsible for assignments	Dr. F.W. Guldenmund	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch English	
Expected prior knowledge	HSM1	
Course Contents	In this course the knowledge from HSM1 will be applied to the actual product that is developed in the minor MedTech Entrepreneurship.	
Study Goals	At the end of the course, participants should be able to 1) Appraise an existing healthcare problem that is to be solved with the intended product 2) Evaluate the impact of the product, both on patient care and on the health system, with a special emphasis on aspects of safety. 3) Explain how the product answers to the adopted problem and how it can be implemented in the health system 4) Present the product consistently and convincingly to actual stakeholders	
Education Method	Tutorial exercises	
Literature and Study Materials	Syllabus health system management 2022	
Assessment	Group assignments consisting of - written report (English) - final presentation	

TBM042A	Finance for Entrepreneurs	4
Module Manager	Dr. T.L. Dolkens	
Instructor	Dr. A. Giga	
Co-responsible for assignments	A. Boru	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is necessary. Knowledge of basic finance and economics helps.	
Course Contents	This course focuses on startup financing. We cover finance tools and concepts that are essential to an entrepreneur when creating a financing strategy. We go over the techniques for setting up and evaluating a financial model and cover the principles of sound financial management. We study fundraising strategy and the entrepreneurial finance industry: the sources of startup funding (e.g. Venture Capital), motivations of entrepreneurs and investors, and negotiations and contracts between them. We also explore latest developments in entrepreneurial finance (e.g. crowdfunding).	
Study Goals	(1) learn how to be a good steward of your business and critically assess financial viability of a business model; (2) know where you can obtain funds for your business, the trade-offs with each funding source, and how it fits your business model; (3) become well-versed in fundraising strategy; (4) master or review fundamental concepts in finance.	
Education Method	The material is illustrated through a combination of lectures, case studies, in-class exercises, and discussions. Depending on the availability and scheduling, we may host guest speakers, either entrepreneurs or investors, to share their experience directly related to the course material.	
Literature and Study Materials	To be announced at the start of the module.	
Assessment	Group project related to the minor project (30%), problem sets (30%), written exam (40%). In case of unforeseen circumstances or due to measures resulting from COVID-19, the exam will take form of a timed take-home exam administered online	
Remarks	If you're not concurrently taking one of the entrepreneurship minors, special permission is needed to attend the class. Please e-mail the instructor for permission.	

TBM043A	MedTech Integration project 1	4
Module Manager	A. Boru	
Instructor	P. Harish	
Instructor	Dr. A. Giga	
Instructor	A. Boru	
Co-responsible for assignments	Dr. T.L. Dolkens	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch English	
Course Contents	This part of the minor MedTech Based Entrepreneurship has the character of a project or internship. During the total integration course (WM4030TU and WM4031TU) the bar is set high, because you always have to get the best out of yourself as an entrepreneur. The self-strengthening cooperation between students and teachers is of great importance in this regard. Self-reliance, proactive behavior and an entrepreneurial attitude contribute to the end result. The intention is that the students prepare themselves to be able to set up their own business or carry out an assignment for a start-up in the MedTech sector. You must literally and figuratively leave the campus. Three process steps form the basis of the integration course: exploring, experimenting and implementing. In the first quarter (WM4030TU) the emphasis is on exploring and experimenting with an eye for implementation. During the second quarter (WM4031TU) the focus shifts to experimenting and implementation, while an open and exploratory attitude remains important. It also offers the space for personal coaching and guidance. Each group of 4 or 5 students will go through the following steps: 1. Consider multiple ideas for your own startup or client 2. Designing and developing a technological innovation 3. Developing business models 4. Testing the ideas and business models in practice (Validation with stakeholders) 5. Prepare a marketing plan 6. Describe the operational plan 7. Making a financial plan 8. Process points 3, 5, 6 and 7 into a business plan 9. Pitching the end result	
Study Goals	The course is supported by lectures that connect to the above subjects. At the end of this course the students are able to do the following: 1. Thinking up different business models 2. Drawing up a business plan 3. Holding a sales and technical pitch 4. Working together as an entrepreneurial team Beside: + Evaluate own work and that of fellow students + Critical analysis and constructive feedback + Critically considering alternative solutions + Orally presenting conclusions and discussing them + Argument and contribute in discussions	
Education Method	This form of project will be supported with lectures and group supervision by coaches with practical experience in the area.	
Assessment	Weekly assignments Final report Oral presentations Weekly progress review (During meetings) Group & Peer evaluations Prior to the meetings, personal reports or assignments must be prepared that will be discussed during the meetings. Group performance can be translated into different individual grades (when there are issues of fairness) In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed final assessment will be modified as follows: - weekly progress meetings will be online - the final (group) presentations will be online the other submissions (weekly assignments, final report etc. are already online so no changes will be made.	

TBM044A	MedTech Integration project 2	4
Module Manager	A. Boru	
Instructor	P. Harish	
Instructor	Dr. A. Giga	
Instructor	A. Boru	
Co-responsible for assignments	Dr. T.L. Dolkens	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch English	
Course Contents	This part of the minor MedTech Based Entrepreneurship has the character of a project or internship. During the total integration course (WM4030TU and WM4031TU) the bar is set high, because you always have to get the best out of yourself as an entrepreneur. The self-strengthening cooperation between students and teachers is of great importance in this regard. Self-reliance, proactive behavior and an entrepreneurial attitude contribute to the end result. The intention is that the students prepare themselves to be able to set up their own business or carry out an assignment for a start-up in the MedTech sector. You must literally and figuratively leave the campus. Three process steps form the basis of the integration course: exploring, experimenting and implementing. In the first quarter (WM4030TU) the emphasis is on exploring and experimenting with an eye for implementation. During the second quarter (WM4031TU) the focus shifts to experimenting and implementation, while an open and exploratory attitude remains important. It also offers the space for personal coaching and guidance. Each group of 4 or 5 students will go through the following steps: 1. Consider multiple ideas for your own startup or client 2. Designing and developing a technological innovation 3. Developing business models 4. Testing the ideas and business models in practice 5. Prepare a marketing plan 6. Describe the operational plan 7. Making a financial plan 8. Process points 3, 5, 6 and 7 into a business plan 9. Pitching the end result	
Study Goals	The course is supported by lectures that connect to the above subjects. At the end of this course the students are able to do the following: 1. Thinking up different business models 2. Drawing up a business plan 3. Holding a sales and technical pitch 4. Working together as an entrepreneurial team Beside: + Evaluate own work and that of fellow students + Critical analysis and constructive feedback + Critically considering alternative solutions + Orally presenting conclusions and discussing them + Argument and contribute in discussions	
Education Method	This form of project will be supported with lectures and group supervision by coaches with practical experience in the area.	
Assessment	Weekly assignments Oral presentations Final report Weekly progress review Group & Peer review Prior to the meetings, personal reports or assignments must be prepared that will be discussed during the meetings. Group performance can be translated into different individual grades (when there are issues of fairness) In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed final assessment will be modified as follows: - weekly progress meetings will be online - the final (group) presentations will be online the other submissions (weekly assignments, final report etc. are already online so no changes will be made.	

A. Boru

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 85757
Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 85757

Dr. T.L. Dolkens

Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89844

Dr. A. Giga

Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89845

Prof.dr.ir. R.H.M. Goossens

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 86340
Room	32.C-3-140

Prof.dr. J. Groeneweg

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
Telephone	+31 15 27 89708

Dr. I. Grossmann

Department	Safety and Security Science
Telephone	+31 15 27 86358

Dr. F.W. Guldenmund

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
Telephone	+31 15 27 82488
Room	31.c1.140

P. Harish

Department	Delft Centre for Entrepreneurship
Department	Delft Centre for Entrepreneurship
Department	Delft Centre for Entrepreneurship

Dr. L. Hartmann

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 88073
Room	31.C2.140

Dr. S. Hinrichs-Krapels

Department	Beleidsanalyse
Telephone	+31 15 27 86390
Room	31.B1.350

Ir. B.R. Joseph

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89831

Mr. J.M. Kooijman

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 81173
Room	31.b2.220

Dr.ing. V.E. Scholten

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89596
Room	31.C2.070

M.T. Sekwenz

Department	Organisatie en Governance
-------------------	---------------------------

Ir. N. Stoimenova

Department	Methodologie en Organisatie van Design
-------------------	--

Unit	Industrieel Ontwerpen
Department	Methodologie en Organisatie van Design

P.B.M. Vandekerckhove

Department	Delft Centre for Entrepreneurship
-------------------	-----------------------------------

Dr. H.G. van der Voort

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 88541
Room	31.b2.140

Dr. G. de Vries

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 86936
Room	31.B2.100

B. Wagner

Department	Organisatie en Governance
Room	31.B2.180

E. Çelik

Department	Delft Centre for Entrepreneurship
-------------------	-----------------------------------
